# Sessional – I (Solution)

**Question 1 – Solution**

**1. Type of Environment**

- **Deterministic vs. Stochastic:** The environment is **stochastic** because the drone's movement can be affected by unpredictable factors like wind, weather changes, and dynamic obstacles (e.g., birds, other drones).
- **Episodic vs. Sequential:** The environment is **sequential** since each decision (such as turning, adjusting altitude, or avoiding obstacles) affects future actions. The drone must continuously consider past states to make better decisions.

**2. Best-Suited Agent Type**

- A **utility-based agent** is the best choice because the drone must not only reach its goal (delivering a package) but also consider multiple factors like shortest path, energy efficiency, and safety.
- A **goal-based agent** would focus only on reaching the delivery destination, but it might not optimize for efficiency and safety.
- A **reflex agent** or **reflex agent with state** would not be ideal because they react based only on current conditions without long-term planning.

**3. Performance Measures**

1. **Delivery Time** – Measures how quickly the package reaches its destination. A lower delivery time improves customer satisfaction.
2. **Energy Efficiency** – Evaluates battery usage and flight path optimization. The drone should complete deliveries using minimal power.
3. **Safety of Navigation** – Ensures the drone avoids collisions with buildings, trees, and other obstacles, reducing risks of failure.

These performance measures help assess the drone's effectiveness and improve its decision-making process.

**Question 2 – Solution**

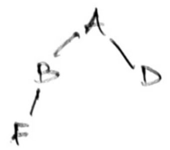
Question 2 :-

Depth first Search :

1



Fringe
A → B
A → D

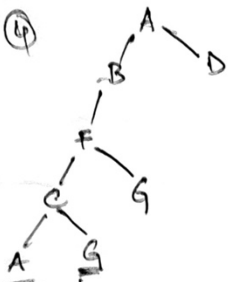
2



Fringe
A → B → F
A → D.

3



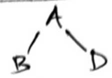
Fringe
A → B → F → C
A → B → F → G
A → D.

4


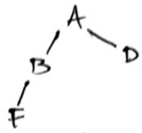

Fringe
A → B → F → C → A
A → B → F → C → G
A → B → F → G.
A → D

↳ Alphabetically "A" will be picked
⇒ Infinite tree
A solution will not be reached.

Breadth first Search :

1



Fringe
A → B
A → D

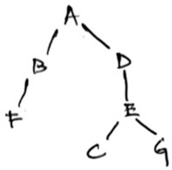
2



Fringe
A → B → F
A → D.

3



Fringe
A → B → F
A → D → E

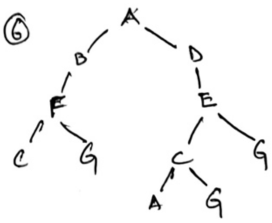
4



Fringe
A → B → F
A → D → E → C
A → D → E → G.

5



Fringe
A → B → F → C
A → B → F → G
A → D → E → C
A → D → E → G.

6



Fringe
A → B → F → C
A → B → F → G
A → D → E → C → A
A → D → E → C → G
A → B → E → G.

G (from E) is picked next which is a Goal state.
⇒ A → D → E → G (Path).

**Question 3 – Solution**

**1. Objective Function for Classroom Assignment**

The objective is to **minimize scheduling conflicts** while ensuring that each class gets a suitable room. A possible function is:

Objective Function = Total Conflicts + Room Capacity Violations + Underutilized Space

where:

- **Total Conflicts (C):** Number of times two classes are assigned to the same room at the same time.
- **Room Capacity Violations (V):** Number of times a class is assigned to a room that is too small.
- **Underutilized Space (U):** Sum of unused seats across all assigned classrooms.

The goal is to **minimize** this function.

**2. Problems with Using Hill Climbing**

Hill Climbing improves schedules step by step but has some challenges:

- **Getting Stuck in Local Optima:** The algorithm may find a schedule that seems good but isn't the best. For example, swapping one class at a time might not fix deeper scheduling conflicts.
- **Plateaus:** If multiple schedules have the same number of conflicts, the algorithm may struggle to decide which way to improve.
- **Ridges:** A better schedule may require temporarily increasing conflicts (e.g., swapping multiple courses), but Hill Climbing avoids such moves.

**3. How Simulated Annealing Helps**

Simulated Annealing (SA) improves scheduling by allowing some bad moves at the start:

- **Escapes Local Optima:** SA can temporarily accept worse schedules, helping it find a better overall arrangement.
- **Explores More Possibilities:** SA makes larger changes early on, then refines the best solution over time.
- **Handles Complex Constraints Better:** SA considers various factors like class size, time slots, and availability without getting stuck easily.

**Conclusion**

Simulated Annealing is more effective because it avoids getting stuck in poor solutions and explores better scheduling options, leading to a more optimized classroom assignment.